

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Solve for the Homogeneous Solution

1. The Characteristic Equation

Consider the *linear homogeneous 2nd-order ODE* written in the form

$$y'' + py' + qy = 0,$$

where p and q are constants. A solution method begins with an *ansatz* (a trial solution) of the form

$$y = e^{rt}$$

where r is an unknown constant to be determined.

Consider the *linear homogeneous 2nd-order ODE with constant coefficients*,

$$y'' + 10y' + 9y = 0.$$

The following tasks uses the *ansatz* method to solve for the general solution.

Task	Show work here
Let $y = e^{rt}$ be the <i>ansatz</i> . Then, determine y' and y'' . —	_____
Show that the ansatz yields the quadratic equation $r^2 + 10r + 9 = 0$ from the ODE. —	_____
Solve the quadratic equation to find values of r . —	_____
State the independent solutions for the ODE, one for each value of r . —	_____
Use the <i>superposition principle</i> to write the general solution.	_____

The choice of $y = e^{rt}$ as the ansatz works because the exponential function has the special property that its derivatives are proportional to itself. The equation $r^2 + pr + q = 0$ is called the *characteristic equation*.

2. Roots of the Characteristic Equation

Using the ansatz $y = e^{rt}$ for the ODE of the form $y'' + py' + qy = 0$ creates the characteristic equation,

$$r^2 + pr + q = 0.$$

The roots of the characteristic equation, determines the form of the general solution:

- If r yields *two distinct real roots*, r_1 and r_2 , the general solution is

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

- If r yields a *repeated real root*, the general solution is

$$y(t) = c_1 e^{rt} + c_2 t e^{rt}.$$

- If r yields *complex conjugate roots*, $r = \alpha \pm \beta i$, the general solution is

$$y(t) = e^{\alpha t} (c_1 \cos(\beta t) + c_2 \sin(\beta t)).$$

For each ODE, state the characteristic equation, find its roots, and write the general solution.

a. $y'' + 3y' + 2y = 0$

c. $y'' + 2y' + 2y = 0$

b. $y'' - 2y' + y = 0$

d. $y'' + 2y = 0$